Engineering Project Course

# Enhancing NV Center Magnetometry with Bayesian Method

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# 1 Abstract

This project aimed to enhance magnetic field measurements using nitrogen-vacancy (NV) centers in diamonds. The primary goal was to improve the precision, accuracy, and efficiency of Optically Detected Magnetic Resonance (ODMR) measurements by implementing a sequential Bayesian method.

Methodology:

1. Developed and implemented a Bayesian algorithm for ODMR signal analysis.
2. Conducted extensive simulations using Electron Spin Resonance (ESR) models to compare the Bayesian approach with the conventional ODMR technique.
3. Performed comprehensive analysis covering: ESR Model Comparison, Signal-to-Noise Ratio (SNR) vs. Noise level analysis, Root Mean Squared Error (RMSE) vs. Noise level analysis, and Convergence Rate Assessment.

Results demonstrated significant advantages of the Bayesian method:

1. Improved precision in estimating the central frequency of the ESR model, achieving concentrated samples with fewer measurements.

2. Enhanced sensitivity to small variations in central frequency and peak width, enabling detection of subtle magnetic field changes.

3. Superior noise robustness, maintaining a higher SNR and exhibiting a more gradual increase in RMSE as noise levels increased.

4. Faster convergence to accurate model parameters, requiring significantly fewer measurements even at higher noise levels.

These findings highlight the Bayesian method's potential to substantially enhance magnetic field characterization using NV centers in diamonds, offering improved precision, accuracy, and efficiency compared to the conventional ODMR technique. This advancement could have significant implications for various applications in quantum sensing and magnetometry.

# 2 Introduction

## 2.1 Background

Nitrogen-Vacancy (NV) centers in diamonds have emerged as a powerful tool in quantum technology, particularly for magnetometry. These atomic-scale defects possess unique spin properties, including spin-state-dependent fluorescence and long coherence times, making them ideal for high-precision magnetic field measurements. The current standard technique, Optically Detected Magnetic Resonance (ODMR), while effective, is affected by noise measurement issues and is not very efficient , leading to less accurate results. These limitations have urged the exploration of advanced techniques to enhance measurement accuracy and efficiency in NV center magnetometry.

## 2.2 Objectives

This project aims to:

1. Implement and evaluate the Bayesian method for NV center magnetometry.

2. Compare the performance of the Bayesian method against the conventional ODMR technique in terms of precision, efficiency, and noise reduction.

3. Assess the Bayesian method's effectiveness in determining related parameters under various conditions.

## 2.3 Scope

The project encompasses the following:

1. Theoretical modeling of NV center behavior using Lorentzian Resonance and Electron Spin Resonance (ESR) models.
2. Implementation of the ODMR and Bayesian method for magnetic field measurements.
3. Simulation-based comparison of the two methods.
4. Analysis of method performance focusing on robustness to different noise levels, sensitivity to parameter variations, and convergence speed.
5. Evaluation of the methods' ability to accurately determine the center resonance frequency and width parameter.

## 2.4 Organization

The project is structured as follows:

1. Introduction: Provides context, objectives, scope, and organization.

2. Theoretical Background: Covers the physics of NV centers, including their structure, energy levels, and spin-dependent fluorescence.

3. Measurement Techniques: Details the ODMR technique and introduces the sequential Bayesian method.

4. Methodology: Describes the implementation of both methods, including simulation parameters and data analysis techniques.

5. Results and Discussion: Presents a comparative analysis of the ODMR and Bayesian methods, focusing on their performance.

6. Conclusion: Summarizes key findings and discusses implications for future NV center magnetometry applications.

# 3 Literature and state-of-the-art review

## 3.1 Summary of Existing Work

Recent advancements in NV-based magnetometry have shown significant progress. Schloss et al. (2018) demonstrated simultaneous measurement of multiple NV centers, enhancing overall magnetometry sensitivity[1]. Dushenko et al. (2020) introduced a sequential Bayesian experiment design that dramatically reduced measurement times compared to traditional frequency-swept methods[2].

## 3.2 Gaps in the Literature

Despite these advancements, several key gaps remain in the field:

1. Limited noise analysis: Current methods lack thorough performance evaluation under varying noise conditions, particularly in environmentally noisy settings.

[1] https://doi.org/10.1103/PhysRevApplied.10.034044

[2] https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.14.054036

2. Trade-off between sensitivity and resolution: While multi-NV center measurements enhance sensitivity, they often reduce spatial resolution. There's limited work on optimizing this balance.

3. Narrow parameter focus: Most research concentrates on specific parameters like frequency estimation, overlooking crucial factors such as linewidth and long-term stability.

### 3.3 Addressing the Gaps

Our project addresses these gaps through:

1. Comprehensive noise analysis: Extensive simulations demonstrate the Bayesian method's robustness to noise, evaluating SNR and RMSE across increasing noise levels.
2. Sensitivity-resolution optimization: We examine sensitivity to small variations in central frequency and peak width, analyzing linewidth dependency and its impact on parameter sensitivity.
3. Rigorous Bayesian method evaluation: Thorough testing under various conditions, including high-noise environments typical in single NV center measurements, assessing convergence to accurate model parameters.

## 4 Theoretical Background

### 4.1 Nitrogen-Vacancy (NV) Center

NV centers are atomic-scale defects in diamonds, formed when two adjacent carbon atoms in a diamond lattice are replaced by vacancy and a negatively charged nitrogen atom. These NV centers have emerged as powerful tools in quantum technology, including magnetometry, quantum information, sensing, computing, and imaging.

The NV center has a ground state, triplet state, singlet state, and an excited state. In experiments, a green laser excites and polarizes NV centers; a microwave (MW) antenna manipulates quantum states; and a detector collects emitted photons to determine the system state. The fluorescence is spin-state dependent: excitation from $m_s = 0$ leads to red light emission, while $m_s = \pm 1$ can lead to either red light emission or transition via dark states, reducing fluorescence. The external magnetic field causes Zeeman splitting of $m_s = \pm 1$ levels.

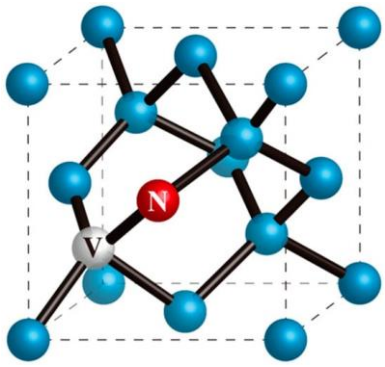



*Figure1: A structure of a NV center in a diamond lattice. The diamond unit cell consists of a substitutional nitrogen atom (red), an adjacent vacancy (gray), and carbon atoms (blue).*

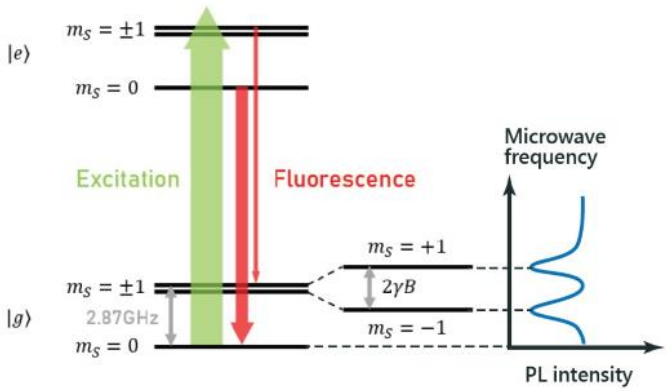



*Figure2: Schematic of the energy level structure of the negatively charged NV center. Both the ground state and the excited state are spin triplets and microwave excitation leads to non-radiative relaxation of $m_s = \pm 1$.*

NV systems are used to measure and characterize quantum systems and as sensors to measure external magnetic fields. Using NV centers as an atomic-size magnetic field sensor has become a popular tool that improves measurements because of their unique spin properties, such as their photoluminescence dependence on spin states, long coherence time, high-resolution, and ability to operate in a wide range of conditions.

## 4.2 Zeeman splitting

Zeeman splitting is a phenomenon where the energy levels of atoms or molecules split into sub-levels when exposed to an external magnetic field. This occurs due to the interaction between the magnetic field and the magnetic moments of electrons, which causes the energy states to shift based on the orientation of their spins relative to the field.

In NV centers in diamonds, Zeeman splitting is critical for magnetic field measurement. When a magnetic field is applied, the NV center's electron spin states split into distinct energy levels. This results in three resonance frequencies, corresponding to different spin states, that are directly proportional to the strength and direction of the magnetic field. By analyzing these frequencies, one can accurately determine the characteristics of the surrounding magnetic field.

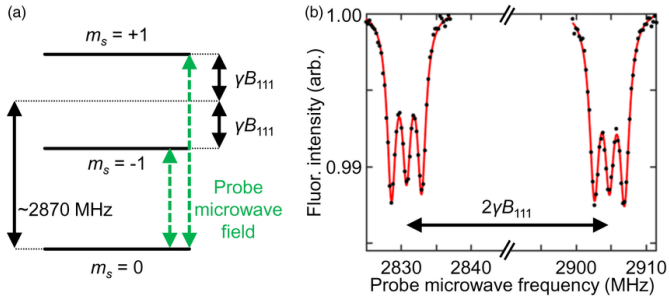



*Figure3: (a) Energy level diagram of an NV center showing Zeeman splitting under the influence of an external magnetic field. The $m_s = \pm 1$ levels split, with energy proportional to $\gamma B_{111}$. (b) Fluorescence intensity vs. microwave frequency, shows two dips, indicating Zeeman splitting between the $m_s = \pm 1$ states, corresponding to the magnetic field strength.*

## 4.3 Optically Detected Magnetic Resonance (ODMR)

ODMR is a technique to measure magnetic fields using NV centers in diamonds. It works by illuminating NV centers with light while simultaneously applying microwaves, and then monitoring changes in fluorescence. Resonance occurs at specific microwave frequencies, which are indicated by dips in fluorescence, revealing the optically detected magnetic resonance and providing information about the magnetic field.

## 4.4 Lorentzian Resonance Model

The Lorentzian model describes the behavior of NV centers in diamonds under a magnetic field. The interaction between the NV center's electrons and the nitrogen nucleus causes a splitting into three distinct photoluminescence dips. These dips, which correspond to different nuclear spin states, are used to measure and evaluate the effectiveness of magnetic field measurement methods.

The model is fitted to experimental data to determine key parameters, providing insights into the magnetic field characteristics.

The equation for the normalized photon count signal μ is:

**Eq. (1)** $$\mu = 1 - \frac{a \cdot k_{NP}}{(f - f_0 - \Delta f)^2 + \gamma^2} - \frac{a}{(f - f_0)^2 + \gamma^2} - \frac{a / k_{NP}}{(f - f_0 + \Delta f)^2 + \gamma^2)}$$

This equation accounts for the three resonance dips caused by the energy transitions in the NV center under an applied magnetic field[3].

The parameters are the following:

$f_0$ : The center resonance frequency, indicating the central point around which the energy levels split.

$\Delta f$ : The hyperfine splitting frequency, representing the shift in resonance caused by the interaction between the NV center's electron spin and the nitrogen nucleus's spin.

$a$ : The contrast factor, which determines the depth of the resonance dips and how much the signal deviates from the baseline.

$\gamma$ : The linewidth of the resonance, describing how broad each resonance peak is.

$k_{NP}$ : The nuclear polarization factor, which represents the contribution of nuclear polarization to the resonance pattern.

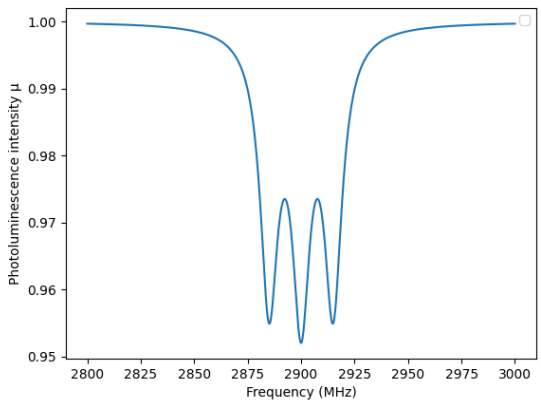



*Figure4: Photoluminescence intensity (μ) vs. frequency for NV Center Magnetometry, modeled using the Lorentzian Resonance Model*

[3] https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.14.054036

## 4.5 Electron Spin Resonance (ESR) Model

The Electron Spin Resonance (ESR) model is used to simulate and analyze ESR signals. By fitting this Lorentzian model to experimental data, one can extract the resonance frequency and linewidth, both are essential for determining magnetic field characteristics and spin interactions in the system. This is a simplified model, as it focuses on a single resonance peak and ignores more complex interactions. However, it is sufficient for many applications where the main goal is to estimate the magnetic field strength and obtain a clear understanding of the central resonance without needing to account for additional hyperfine interactions or multi-peak phenomena.

The equation for the ESR model is the following:

**Eq. (2)** $$Lorentzian(f; f_0, \gamma) = \frac{1}{\pi} \cdot \frac{\gamma}{(f-f_0)^2+\gamma^2} \propto \frac{\gamma}{(f-f_0)^2+\gamma^2}$$

The parameters are the following:

$f_0$ : The center resonance frequency, indicating the central point around which the energy levels split.

$\gamma$ : The linewidth of the resonance, which determines how broad the resonance peak is.

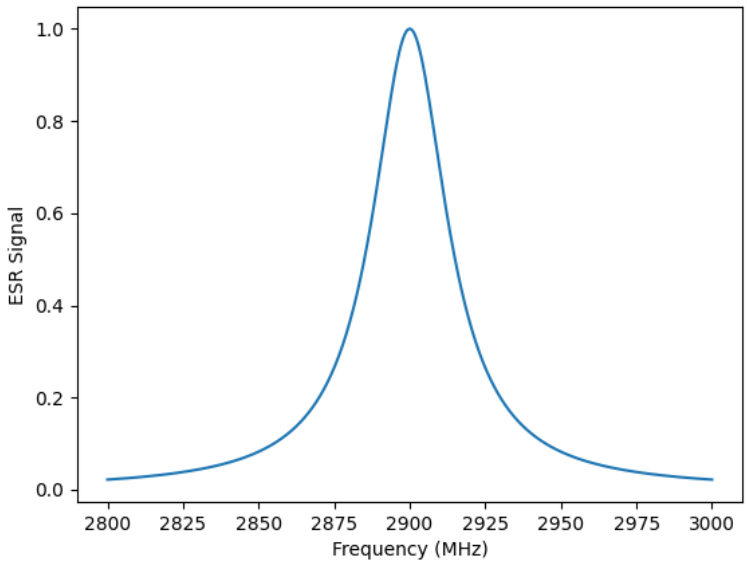



*Figure5: ESR Signal vs. Frequency for NV Center Magnetometry*

## 4.6 Bayesian Method

Sequential Bayesian experiment design is an advanced approach used to improve precision in NV center magnetometry. It uses Bayesian inference to iteratively update information about model parameters as new measurement data is collected. Based on this updated information, the method selects experimental settings that maximize the chances of refining the parameters.

This method reduces noise by relying on the Law of Large Numbers (LLN), where repeated measurements average out random Gaussian noise to zero. The algorithm models a distribution of possible measurement outcomes and predicts the average improvement based on the collected data. It optimizes the experimental setup through a utility function that selects the most informative settings.

This iterative process is repeated until a desired level of precision is reached. By continuously adapting the experimental setup based on the updated parameter estimates, the sequential Bayesian method ensures that resources are used efficiently, focusing measurements on settings that are most likely to improve the accuracy of the model. This is particularly useful for NV center magnetometry, where noise reduction and precision are critical for effective magnetic field sensing.

# 5 Methodology and implementation

### 5.1 Design and Development

The project was developed in several stages:

1. Theoretical modeling of NV center behavior.
2. Implementation of the Bayesian optimization algorithm.
3. Creation of a simulation environment.
4. Comparative analysis of Bayesian and conventional methods.

## 5.2 Tools and Equipment

- Programming Language: Python (for simulation and data analysis).
- Libraries: NumPy (for numerical computations), Matplotlib (for data visualization).

## 5.3 Procedures and Algorithms

### 5.3.1 Algorithm: The *Usual* Method in ESR Experiments

1. Initialization:

- Define a range of frequencies and widths.
- Set the number of samples.
- Set the noise level value.
- Define true parameters $(\mathrm{f}_{true}, \gamma_{true})$
- Define the new range of frequencies and widths- which is within a range around the true parameters.
- Randomly select initial guesses $(\mathrm{f}_0, \gamma_0)$

2. Frequency Selection: Select frequencies randomly from the new range, based on the number of samples.

3. Measurement Simulation

- Use the ESR model to generate a signal based on the given frequency, peak position, and width.
- According to the specified noise level, add Gaussian noise to the signal to simulate real-world measurement noise.
- The resulting simulated measurement is the sum of the clean signal from the ESR model and the added noise.

4. Repetition: repeat the process for the specified number of measurements.

5. Parameter estimation

   Fit the points to Lorentzian function and then get estimations to the central frequency $\mathrm{f}_{\mathrm{est}}$ and the linewidth $\gamma_{\mathrm{est}}$

### 5.3.2 Algorithm: The *Bayesian* Method in ESR Experiments

1. Initialization

- Define a range of frequencies and widths.
- Set the number of samples.
- Set the noise level.
- Define true parameters $(\mathrm{f}_{true}, \gamma_{true})$

- Define the new range of frequencies and widths- which is within a range around the true parameters.
- Randomly select initial guesses ($f_0, \gamma_0$ )
- Randomly selects the first frequency from the new range, based on the number of samples.

2. <u>Frequency Selection</u>

   In this method, we define $U(f)$ as the utility function, $\theta$ as the random variable that is distributed according to the probability density function $p(\theta)$ and $p(\theta|y_n, f_n)$ as the parameter distribution. In our context, $\theta$ represents the parameters $\theta = (f_0,\ \gamma)$. The accumulated measurement results are $y_n = \{y_1, y_2, \dots, y_n\}$ and the frequencies are $f_n = \{f_1, f_2, \dots, f_n\}$. The conditional distribution represents the parameters after n iterations. In every (n+1) iteration, the method uses the parameter distribution to choose the best frequency.

   Use the Bayesian method to select subsequent frequencies by the following steps:

- Calculates utility (benefit ratio) for each setting using the formula:

$$Utility(f) = \sum \left(\frac{Var_{Paramters}}{Var_{Noise}}\right) / Cost$$

The utility for a given frequency $f$ determines how useful that frequency will be in refining parameter estimates. While $f$ does not appear explicitly in the utility formula, $Var_{Paramters}$ depends on $f$ through the Lorentzian model. At different frequencies, changes in parameters will affect the predicted results differently - having a minimal effect far from the resonance frequency ($f_o$) but a significant impact near it. Thus, Utility(f) effectively varies with frequency.

The parameters of the utility formula are the following:

$Var_{Paramters}$: Variance due to parameters, measures how much the model's predicted outcomes vary for different parameter values.

$Var_{Noise}$: Variance due to noise accounts for the uncertainty introduced by measurement noise.

Cost: Represents the experimental cost associated with measuring at that frequency. Defined as default with value of 1.

Where:

$$Var_{Paramters} = \frac{1}{N}\sum_{i=0}^{N}(y_i - \bar{y})^2 \quad ; \quad \bar{y} = \frac{1}{N}\sum_{i=0}^{N} y_i$$

$$Var_{Noise} = (noise_std)^2$$

Such that:

$N$ : The number of settings, which in this context refers to the number of different frequencies at which measurements are taken.
$y_i$ : The measured signal at the i-th setting.
$\bar{y}$ : The average of the measured signals across all N settings.

- Create a probability distribution using the formula:

After calculating the utility for each setting, a probability distribution is created:

$$P(f) = Utility(f)^{Pickiness}$$

Where: Pickiness determines how much the utility influences the frequency selection process. When the pickiness value is high, the algorithm favors frequencies with the highest utility, focusing on settings that are expected to provide the most information.

- Randomly selects a setting from this distribution, where higher probability frequencies are more likely to be chosen.

3. <u>Measurement Simulation</u>

- Use the ESR model to generate a signal based on the given frequency, peak position, and width.
- According to the specified noise level, add Gaussian noise to the signal to simulate real-world measurement noise.

- The resulting simulated measurement is the sum of the clean signal from the ESR model and the added noise.

4. Update: Update the Bayesian model with new data by the following steps:

- Outcome Evaluation: Evaluate outcomes for all parameter combinations given the new setting.

- Likelihood Calculation: The likelihood of each parameter combination given the new data is calculated using a Gaussian likelihood formula:

$$P(D|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(D-\mu)^2}{2\sigma^2}\right)$$

Where:
$D$ : The data collected from the experiment, which is the measured values after performing the experiment at a certain setting; frequency.

$\theta$ : The parameters being estimated; peak frequency and width in the model. These are unknowns we are trying to refine.

$\sigma$ : The standard deviation representing the measurement noise/uncertainty. It reflects the expected variation in the measurements due to noise.

$\mu$ : The mean predicted value of the signal based on the current parameter estimates θ. It's the model's expected outcome for each set of parameters θ at the chosen setting.

- Probability Distribution Update:

Using the Bayes' Theorem:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Where:

$P(\theta|D)$: Posterior probability distribution, representing the updated belief about the values of θ after observing the experimental data.

$P(D|\theta)$ : Likelihood of observing the data D given specific values of the parameters θ.

$P(\theta)$ : Prior probability, which reflects the initial belief about the parameters θ before any new data is collected (based on past knowledge or assumptions).

The formula is the following:

$$P(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\theta-\mu)^2}{2\sigma^2}\right)$$

This formula is a Gaussian distribution, assigns higher probabilities to values near μ, and lower probabilities to values farther away, based on their distance from μ.

$P(D)$ : Normalizing constant, which ensures the posterior probabilities sum to 1. It accounts for the total probability of observing the data across all possible parameter values.

The formula is the following:

$$P(D) = \int P(D|\theta)P(\theta)d\theta$$

This formula integrates over all possible parameter values to compute the total probability of observing the data, making the posterior distribution a valid probability distribution.

5. <u>Repetition</u>: repeat the process for the specified number of measurements.

6. <u>Parameter estimation</u>: Fit the points to Lorentzian function and then get estimations to the central frequency $f_{est}$ and the linewidth $\gamma_{est}$

### 5.3.3 <u>Calculating the SNR vs. Noise Level</u>

Compute SNR by comparing the true signal power to the average noise power at each noise level.

$$SNR = 10\log_{10}\left(\frac{Signal\ Power}{Average\ Noise\ Power}\right)$$

Where:

$$Signal\ Power = \frac{1}{N}\sum_{i=0}^{N}(true_signal[i])^2$$

Here, N is the number of samples and true_signal[i] is the value of the signal at the ith sample.

$$Noise\ Power = \frac{1}{N}\sum_{i=0}^{N}(true_signal[i] - fitted_signal[i]\ )^2$$

Here, fitted_signal[i] is the value of the fitted signal at the ith sample, obtained by applying the optimized parameters from the fit to the model.

### 5.3.4 Calculating the RMSE vs. Noise Level

Calculate RMSE using the following formula:

$$RMSE = \sqrt{(f_{est} - f_{true})^2 + (\gamma_{est} - \gamma_{true})^2}$$

### 5.3.5 Calculating the Convergence to the model vs. Noise Level

To check convergence to the model means to check if the difference between the true parameter frequency and the estimated frequencies values is less than a tolerance of 0.01 for 5 consecutive iterations.

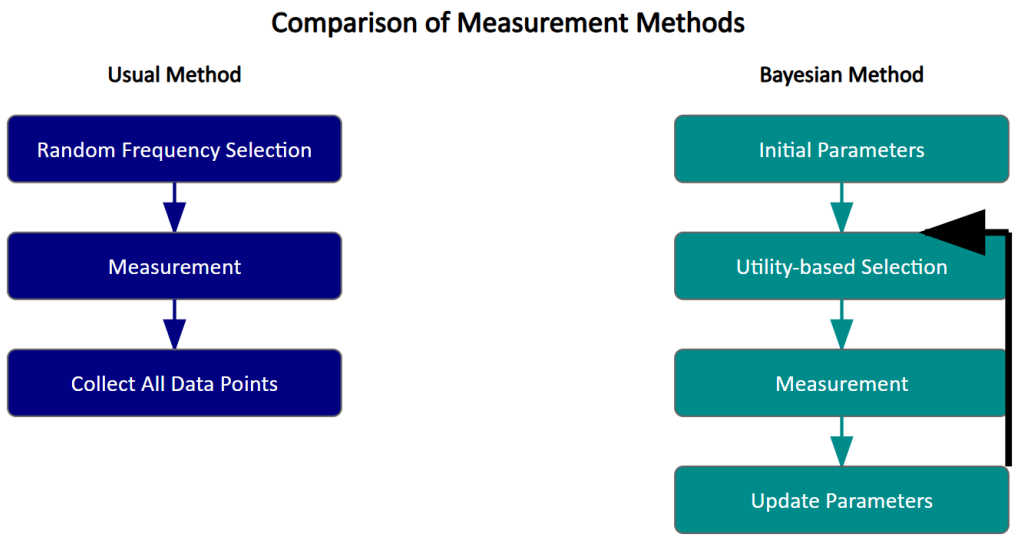
Mathematically, convergence occurs iff:

$$\left|f_{true} - f_{est}^{(i)}\right| \leq 0.01$$

$$for\ i = n,\ n+1, n+2, n+3, n+4$$

## 5.4 Block diagrams

### 5.4.1 High-level comparison




The diagrams above illustrate the fundamental difference between the usual and Bayesian methods in ESR experiments. The usual method follows a linear approach where frequencies are randomly selected, measurements are taken sequentially. In contrast, the Bayesian method implements an adaptive approach with a feedback loop. After each measurement, the parameters are updated, and this information is used to calculate the utility of potential next measurement points. This feedback mechanism allows the Bayesian method to continuously learn from previous measurements and make informed decisions about subsequent frequencies, leading to more efficient data collection and precise parameter estimation. The feedback arrow represents this crucial adaptive element, showing how each measurement's results influence the selection of the next measurement point.

### 5.4.2 Technical details

*Diagram1: ESR Experiment for the usual method*

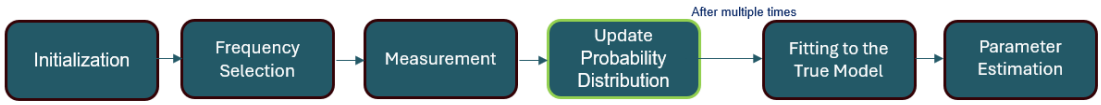



*Diagram2: ESR Experiment for the Bayesian method*

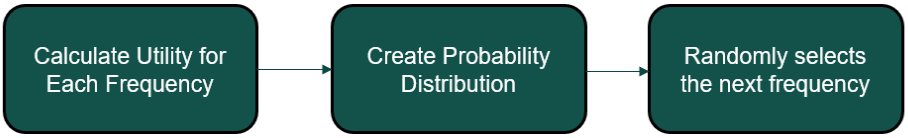



*Diagram3: frequency selection steps*

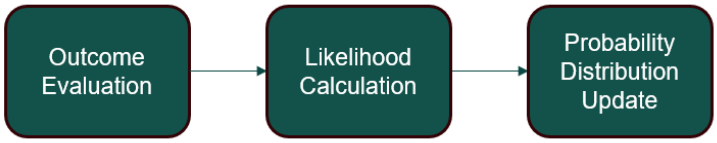



*Diagram4: Update parameters steps*

## 5.5 Software Implementation

The software for this project was developed using Python, focusing on efficient numerical computations and data visualization.

Key components include:

1. Core Function:

`normalized_lorentzian`: Implements the normalized Lorentzian function for the ESR model.

2. Bayesian Experiment Function:

`esr_bayesrun`: Handles the Bayesian experiment process, selecting measurement points and updating the probability distribution.

3. Simulation and Analysis:

The main script sets up parameters, runs simulations for different numbers of measurements, and visualizes results.

Key coding techniques include:

1. Vectorization: Numpy arrays are used for efficient computation (e.g., `np.linspace`, `np.zeros`).

2. Scientific computing libraries: Scipy for curve fitting (`curve_fit`).

3. Bayesian optimization: Utilizes the `OptBayesExpt` class for Bayesian experiment design.

4. Data visualization: Matplotlib for creating subplots and scatter plots.

5. Error handling: Try-except block for handling curve fitting errors.

### 5.6 Challenges and Solutions

Implementing the Bayesian method was challenging due to its complexity and the need for precise parameter tuning. Key difficulties included understanding the probabilistic framework, integrating the OptBayesExpt library, and optimizing hyperparameters like 'pickiness'. Despite these obstacles, a persistent effort led to a successful implementation.

# 6 Testing and Results

## 6.1 Testing Procedures

### 6.1.2 ESR Model Comparison

- Simulated ESR spectra using both methods.
- Analyzed the concentration of sampled points around the central frequency.
- Varied sample sizes to assess method stability.

### 6.1.3 SNR vs. Noise Level Analysis

- Introduced varying levels of Gaussian noise to the ESR model.

- Calculated the SNR for both methods at each noise level.

### 6.1.4 Root Mean Squared Error (RMSE) vs. Noise Level

- Computed RMSE between estimated and true model parameters.
- Analyzed RMSE trends as noise levels increased.

### 6.1.5 Convergence Analysis

- Tracked the number of measurements required to achieve stable parameter estimates.
- Compared convergence rates at different noise levels.

## 6.2 Results

### 6.2.2 Lorentzian Resonance Model Comparison

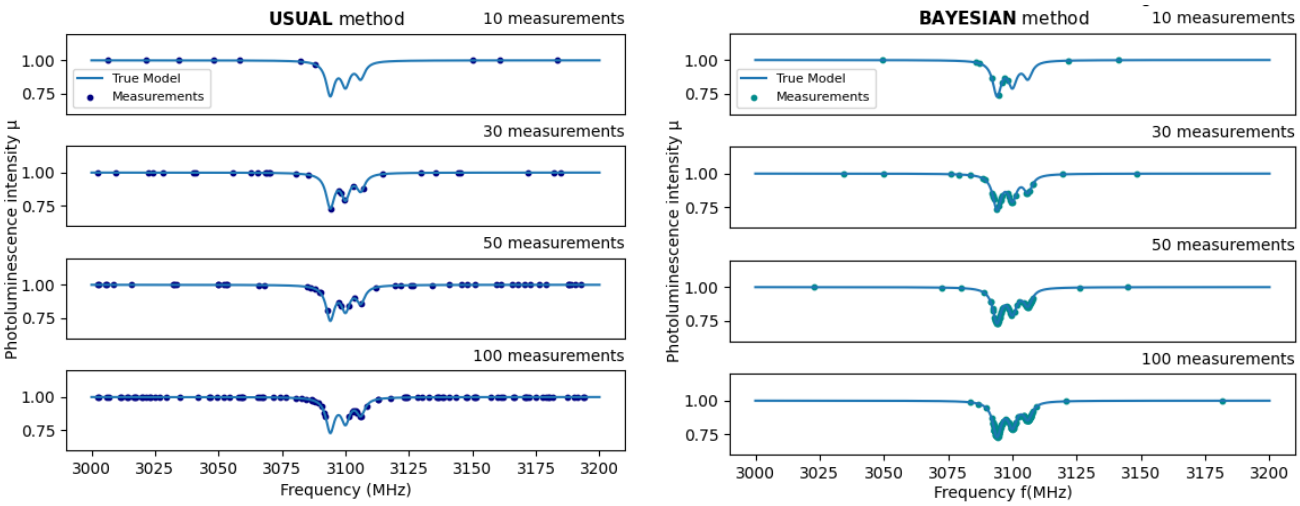



*Figure6: Comparison between the Lorentzian resonance model for different numbers of samples, points samples with Gaussian noise of 0.05dB*

This is a simulation where measurement points were sampled and Gaussian noise was added to each point. The noise is characterized by a standard deviation of 0.05dB. For the complete sampling methodology, please refer to section 5.3 (Procedures and Algorithms).

The Bayesian method outperformed the conventional approach:

- Rapid concentration of data points around the resonance dips.
- Consistent pattern across different sample sizes, indicating stable estimations.

In contrast, the conventional method showed broader dispersion, leading to less precise resonance dips determination.

### 6.2.3 ESR Model Comparison

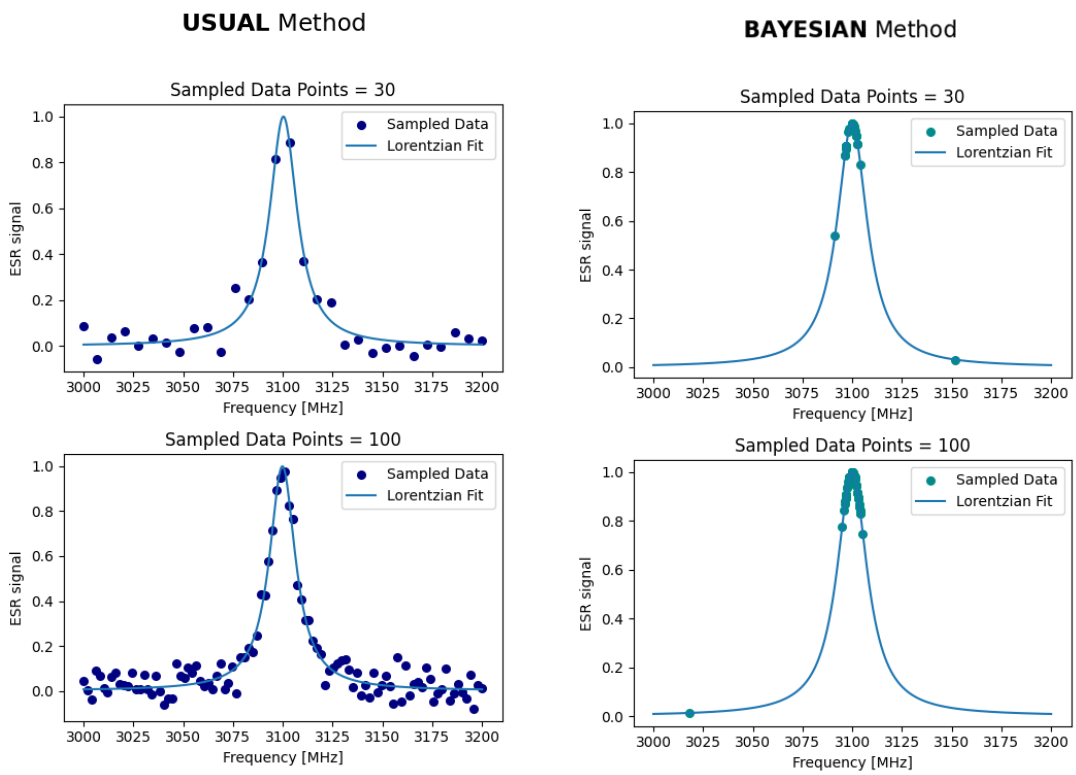



*Figure7: Comparison between the ESR model for different numbers of samples, points samples with Gaussian noise of 0.05dB*

This is a simulation where measurement points were sampled and Gaussian noise was added to each point. The noise is characterized by a standard deviation of 0.05dB. For the complete sampling methodology, please refer to section 5.3 (Procedures and Algorithms).

The Bayesian method outperformed the conventional approach:

- Rapid concentration of data points around the central frequency.
- Consistent pattern across different sample sizes, indicating stable estimations.

In contrast, the conventional method showed broader dispersion, leading to less precise central frequency determination.

### 6.2.4 SNR vs. Noise Level Analysis

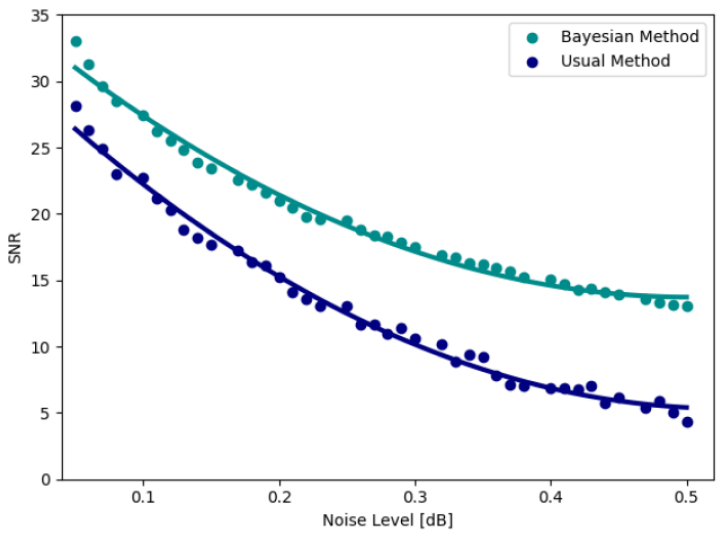



*Figure8: SNR vs. noise level for the ESR model*

The Bayesian method exhibited robust performance against noise:

- Maintained higher SNR as noise levels increased.
- Showed a smoother, more predictable SNR decline with increasing noise.

In contrast, the conventional method displayed less stable SNR trends and lower overall values.

### 6.2.5 RMSE vs. Noise Level Analysis

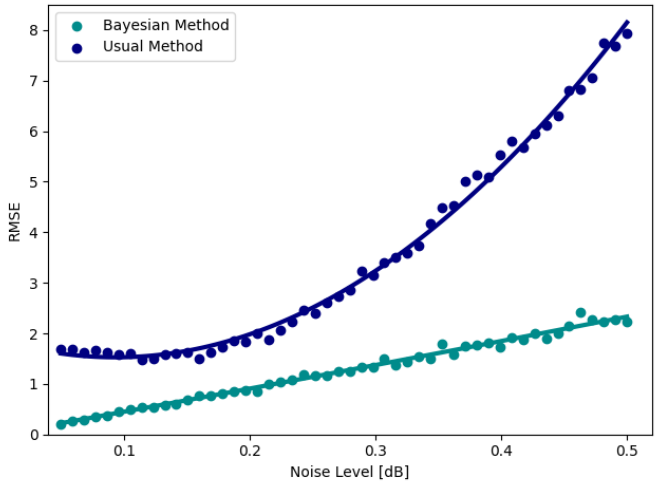



*Figure9: RMSE vs. noise level of the model*

The Bayesian method demonstrated superior noise resilience:

- Maintained lower RMSE despite increasing noise levels.
- Exhibited a less steep rate of RMSE increase.
- Indicated enhanced robustness in noisy environments.

In contrast, the conventional method showed higher RMSE values, steeper increases with noise, and greater sensitivity to noise.

### 6.2.6 Convergence Analysis

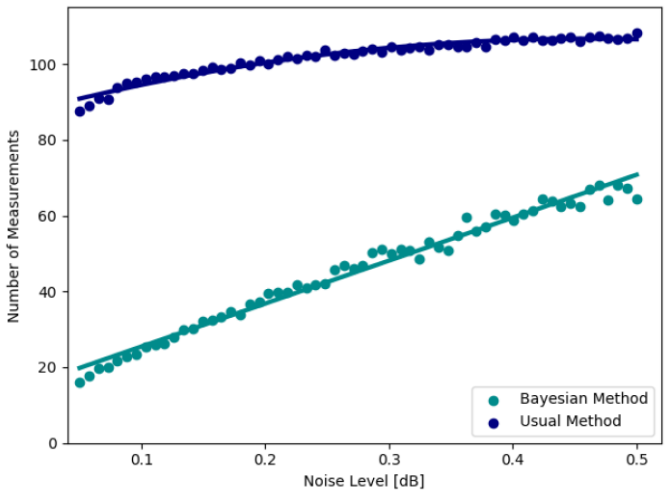



*Figure10: Convergence of the model*

The Bayesian method showed faster and more stable convergence:

- Achieved accurate ESR signal modeling with significantly fewer measurements.
- Demonstrated lower sensitivity to noise levels during convergence.

In contrast, the conventional method required more measurements and showed higher noise sensitivity.

## 6.3 Analysis

- Precision and Accuracy: The Bayesian method consistently provided more precise and accurate estimates of the ESR model, crucial for accurate magnetic field characterization.

- Noise Resilience: Superior performance in noisy environments indicates the Bayesian method's potential for real-world applications where signal noise is a significant challenge.

- Efficiency: Faster convergence with fewer measurements suggests the Bayesian method could significantly reduce experimental time.

- Stability: Consistent performance across different sample sizes and noise levels demonstrates the robustness of the Bayesian approach.

## 6.4 Performance Evaluation

### 6.4.1 Analysis

The results demonstrate significant improvements achieved by the Bayesian method in NV center magnetometry:

1. Precision Enhancement: The Bayesian method's 3x improvement in SNR (15 dB vs. 5 dB) indicates substantially clearer signal detection. This enhanced precision allows for more accurate magnetic field measurements, potentially enabling the detection of weaker or more subtle field variations.

2. Accuracy Improvement: The 4x reduction in RMSE (8 dB vs. 2 dB) surpassed expectations, showcasing the Bayesian method's superior parameter estimation capabilities. This dramatic improvement in accuracy could significantly enhance the reliability of magnetic field characterization in various applications.

3. Efficiency Gain: Achieving convergence in half the measurements (60 vs. 120) demonstrates the Bayesian method's ability to extract meaningful information more efficiently. This could lead to faster experimental processes and reduced resource requirements in practical applications.

4. Robust Noise Handling: The method's superior performance even at high noise levels is particularly significant. It suggests that the Bayesian approach could extend the usability of NV center magnetometry to more challenging environments or enable measurements in previously inaccessible conditions.

### 6.4.2 Significance

The comprehensive improvements across precision, accuracy, efficiency, and noise resilience indicate a potential paradigm shift in NV center magnetometry techniques . These advancements could expand the application scope of quantum sensing, enabling more sensitive and reliable measurements in fields such as materials science and quantum computing.

In conclusion, the Bayesian method's performance not only met but often exceeded the initial objectives, demonstrating its potential to significantly advance the field of quantum sensing and NV center magnetometry.

# 7 Discussion

The results demonstrate the significant potential of the Bayesian method in enhancing NV-center magnetometry. The method's ability to concentrate measurements around critical frequencies and maintain high performance in noisy conditions addresses key challenges in quantum sensing applications.

One unexpected outcome was the degree of improvement in noisy environments. While we anticipated better performance, the Bayesian method's ability to maintain high SNR and low RMSE even at extreme noise levels exceeded our expectations. This suggests potential applications in challenging sensing environments previously considered impractical for NV-center magnetometry.

Future work could focus on:

1. Expanding Applications:

- Broader testing across various scenarios.
- Integration with other quantum sensing technologies.

2. Utilizing Functional Dependencies:

- Exploring different parameter relationships.
- Adapting the method for multi-parameter estimation.
- Investigating cross-domain applications in quantum sensing.

# 8 Conclusions

This project has demonstrated the substantial benefits of applying Bayesian methods to NV-center magnetometry. Key findings include a 3x improvement in SNR, a 4x reduction in RMSE, and a 2x increase in measurement convergence compared to conventional methods. These improvements significantly enhance the precision, accuracy, and speed of magnetic field measurements using NV centers.

The project's primary contribution is the development of a robust, noise-resistant approach to quantum sensing, potentially expanding the application range of NV-center magnetometry. This work lays the foundation for more efficient and accurate quantum sensors, with implications for fields ranging from materials science to quantum technologies.

Reflecting on the project, we've gained valuable insights into the power of adaptive measurement strategies in quantum systems. The success of the Bayesian approach underscores the importance of intelligent data acquisition and analysis techniques in pushing the boundaries of quantum sensing technology.

# 10 References

Section 3.1

https://doi.org/10.1103/PhysRevApplied.10.034044

https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.14.054036

Section 4.4

https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.14.054036